Abstract

Now more than ever, the protection of personal data and vital infrastructure from cyber-attacks is crucial in

our digital world. With the cyber threat landscape constantly changing, traditional intrusion detection systems battle to keep up with both the technical advancements and complexity of todays's attacks. In particular, signature-based systems are notoriously bad at detecting zero-day exploits and have a high false positive rate.

In anticipation to the growing demand for solutions that are both more scalable and fault-tolerant, this research attempts to investigate whether deep neural networks can be useful in the area of intrusion detection. Using machine learning, mainly deep learning architectures as a tool to create intrusion detection systems that can more efficiently detect and prevent each kind of cyber-attacks.

This report describes our journey in this space — building and evaluating such systems. Building a model- Each step right from defining the problem statement to handling preprocessing of datasets, developing and training models under such load and finally assessing those performances is noted down meticulously. In addition, better understanding or the insights from visualization of how model training really works (training process) could also provide better perspective on whether the developed models meet expectations so far.

We hope that our research can provide insights and methodologies for other works on intrusion detection, it will contribute to develop the first line of defense against cyber-attack in different sectors. We hope to bring unprecedented resilience to important systems and secure the digital foundation of modern society by unleashing the power of deep neural networks.

Aman Jawalekar

22070243022

MSc Data Science & Spatial Analytics